# Antonio López Muzás

Senior Software Engineer

Spain · remote

lopezmuzas@gmail.com · linkedin.com/in/antoniolopezmuzas · github.com/lopezmuzas · lopezmuzas.github.io/resume

## I build business software for regulated sectors, and the plumbing that lets AI agents work with it safely.

**Senior software engineer with 19 years of experience** building backend and full-stack systems in Python, PHP and TypeScript, with PostgreSQL, Docker and AI/LLM integrations, for vehicle inspection, energy and healthcare.

I start from the business: I improve how a company works through clean architecture and sound engineering practice.

Most of my career has been in vehicle inspection, energy and healthcare: systems that run a business day to day, where a wrong number has consequences. Today that means products for inspection stations and energy retail, and AI features with hard limits on what the model can touch. In my spare time I build a home run by agents, tools for my research and the servers they run on. Alongside, I'm preparing a PhD proposal on federated learning over data spaces.

I have extensive experience working remotely with multidisciplinary teams in startups, research departments and small and medium-sized businesses.

## Experience

`Mar 2020 – Present`

### Software Engineer · Ibatec

Vehicle inspection (ITV) and energy · remote

I design and build the software that inspection stations and an energy sales network run on, end to end: from the database to the phone line.

- Station management, ERP and an online booking system that handles thousands of appointments a week, plus training campus and HR tools and integrations with suppliers and inspection equipment.
- The AI layer: a phone agent that books appointments, a checker for vehicle-modification files, natural-language analytics over inspection data and licence-plate recognition.
- An electricity and gas sales platform that handles tens of thousands of contracts a year, from offer to activation with each retailer.
- Data sharing between stations and in agrifood on Ocean Protocol data spaces.

Technologies: Python, FastAPI, PHP, Laravel, React, PostgreSQL, MCP, LLMs

`Mar 2015 – Feb 2020`

### Software Engineer · Optretina

Healthcare · Barcelona area, on site and remote

I built a remote retinal screening platform that reviews more than 1,000 people a day and connects them with retina specialists.

- The screening platform, on the Slim PHP framework.
- APIs that connect the platform with deep-learning models that flag retinal conditions.
- Worked directly with partners such as Zeiss, Sanitas and Clínica Baviera, in an international team.

Technologies: PHP, Slim, REST APIs, Deep learning, MySQL

Feb 2014 – Feb 2015

### Freelance Full-stack Developer

Madrid · remote

Cloud web applications for small businesses: corporate websites, a crowdfunding platform for social projects and an iPhone app for splitting payments among friends.

Aug 2011 – Feb 2014

### Backend Developer · Ondango

Berlin, Germany · on site and remote

Built an e-commerce platform that ran inside Facebook, focused on a smooth shopping experience. When Facebook changed its policies, the company pivoted to a fashion marketplace and the backend went with it.

Technologies: CakePHP, Facebook Open Graph, AWS

Mar 2007 – Jan 2011

### Web Developer · Planes Asesoría y Soluciones Informáticas

Barbastro, Spain · on site

Web solutions for small and medium-sized companies: online shops on an in-house PHP framework, a Java web app for medical appointments and a billing system in Visual Basic 6.

## Selected work

### At Ibatec

#### Phone booking agent

A voice agent answers the station's phone and books, moves or cancels inspections. It works through MCP tools on top of the booking system, so it can only do what the booking rules allow.

Technologies: FastAPI, PostgreSQL, MCP, Voice agents

#### Vehicle-modification file checker

Reads the PDF dossier of a modified vehicle with a multimodal model, checks it against the regulation with deterministic rules and only then asks an LLM to review what the rules can't settle.

Technologies: Python, Gemini, pydantic, Rules engine

#### Ask the inspection data

Questions in plain language become SQL through a curated semantic catalogue. The model only writes the query: it is validated with sqlglot and run read-only on a nightly DuckDB/Parquet export.

Technologies: NL2SQL, sqlglot, DuckDB, Streamlit

#### Plates, cards and certificates

Licence-plate detection with a CNN and OCR, and text extraction from inspection cards and certificates of conformity.

Technologies: TensorFlow/Keras, OpenCV, Tesseract, Cloud Vision

#### Brake-tester bridge

Connects the inspection software with a brake-test bench over its proprietary protocol, exposed as WebSocket and REST and shipped as a Windows executable.

Technologies: Python, WebSockets, PyInstaller

### Energy contracts, from offer to activation

A platform for an electricity and gas sales network: customers and their supply points, rates and products from several retailers with a best-option comparison, bulk contract uploads and a daily processing queue, for tens of thousands of contracts a year. Each contract moves through the retailer's API, webhooks or file exchange, with per-retailer discard rules and a signature record.

Technologies: Laravel, Livewire, OAuth (Passport), MariaDB, Webhooks

### Agrifood data space

An Ocean Protocol node, a custom marketplace and a Compute-to-Data algorithm that benchmarks agricultural-machinery inspections without the data leaving its owner.

Technologies: Ocean Protocol, Next.js, Ocean.js, Python

## Side projects

### A house run by one agent

A single agent for the whole home coordinates sub-agents for cooking, energy and cleaning, so each can use what the others know: recipes and meal history, hourly electricity prices, smart plugs and the robot vacuum on a 3D floor plan. Devices are read locally over MQTT, and the home's knowledge lives in its own database, independent of any manufacturer or AI provider.

Technologies: FastAPI, MCP, MQTT, Three.js, Raspberry Pi

### A reading app that moves my research forward

Every paper I read has to produce something: a claim, a gap, an experiment idea or an explicit discard. It works on top of Zotero, has its own phone reader (highlight, annotate, translate and resume at the same paragraph on any device) and writes the research log by itself. Claude organises the reading queue over MCP.

Technologies: FastAPI, React, pdf.js, Zotero, MCP

### Local-first AI gateway

A Claude-compatible API that answers with a small local model by default and routes to Anthropic, OpenAI, Gemini or Mistral only where the data is allowed to leave.

Technologies: FastAPI, Ollama, Multi-provider

### Servers that agents can run

One standard for provisioning, deploys, backups and recovery across a small VPS fleet, and a monitoring app on Prometheus, Loki and Grafana that agents can operate over MCP, without SSH.

Technologies: Docker, nginx, restic, Prometheus, Loki

### A work log for coding agents

An MCP service, a small Go sensor and a Claude Code plugin that record what each agent session did, decided and left open, so the next session picks up where the last one stopped.

Technologies: MCP, Go, Claude Code

# Research

### Federated learning over sovereign data spaces

In Compute-to-Data each step is an isolated, stateless and paid job, which makes multi-round federated training costly. My work measures that cost and proposes verifiable one-pass model evaluation.

Technologies: Federated learning, Ocean Protocol, Gaia-X, Compute-to-Data

**Nodum**
The lab for this research: Ocean nodes that deploy from a clean server in one command, federated averaging over Compute-to-Data, a marketplace rewritten with DDD, and synthetic datasets and algorithms ready to publish.

Technologies: Python, FastAPI, web3.py, NumPy, PyTorch

**Interacting with Linked Data through an Automatic Information Architecture.** Conference paper. Read

**Publishing and Interacting with Linked Data.** Conference paper. Read

## Skills

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|---|---|
| **Languages** | Python, PHP, TypeScript and JavaScript, SQL, Bash; some Go |
| **Backend** | FastAPI, SQLAlchemy (async), Alembic, pydantic · Laravel, Slim, Symfony · Flask, Django |
| **Frontend** | React, Vite, Inertia, Tailwind · Livewire, Next.js, Three.js |
| **Data** | PostgreSQL (row-level security, pgvector), MariaDB/MySQL, DuckDB and Parquet, Qdrant, Redis, MongoDB, Elasticsearch |
| **AI & ML** | MCP · Claude, OpenAI, Gemini, Mistral, Ollama · RAG (hybrid search, reranking, evaluation) · NL2SQL · voice agents · PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, OpenCV · federated learning |
| **Data spaces** | Ocean Protocol (nodes, Compute-to-Data, Ocean.js), Gaia-X/Pontus-X, ODRL, web3.py |
| **Infra & ops** | Docker and Compose, nginx, Let's Encrypt, GitHub Actions, Make · Prometheus, Loki, Grafana, OpenTelemetry · restic · AWS, DigitalOcean |
| **Testing & quality** | pytest, PHPUnit, Pest · static analysis (pyright, PHPStan) · integration tests against real databases · code review and security review · CI/CD with GitHub Actions |
| **Messaging** | RabbitMQ, MQTT, Apache Kafka |
| **Architecture** | DDD, hexagonal architecture, CQRS, event sourcing, multi-tenancy |

## Education

2009 – 2011

**MSc in Human-Computer Interaction** · Universitat de Lleida and DERI Galway

Thesis *Interacting with the Semantic Web*, written during a funded internship at DERI Galway, Ireland. Specialised in information architecture, accessibility and user-centred design, and contributed to the publications of the GRIHO research group.

2003 – 2009

**BSc in Computer Science** · Universitat de Lleida

Final project: visualisation plugins for the Rhizomer Semantic Web platform. Thesis · Slides (in Spanish)

## Languages

Spanish (native) · English (professional working proficiency, C1) · Russian (intermediate, conversational)